Math 540

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Week 7 Problems

Exercise 1. *Suppose that S, T are bounded linear operators on a Hilbert space H. Suppose ST is compact. Must either S or T be compact? Either prove or give an explicit counter-example using $H = \ell^2$.*

Proof. Define $S : \ell^2 \to \ell^2$ by:

$$(x_1, x_2, x_3, x_4, ...) \mapsto (x_1, 0, x_2, 0, ...).$$

Define $T : \ell^2 \to \ell^2$ by:

$$(x_1, x_2, x_3, x_4, ...) \mapsto (0, x_2, 0, x_3, ...).$$

Neither S nor T are compact, but $S \circ T$ is compact as it is the zero operator. □

Exercise 4. *Let (λ_n) be a bounded sequence of complex numbers. Show that the bounded linear operator $T : \ell^2 \to \ell^2$ defined by:*

$$Te^n = \lambda_n e^n$$

(and extended linearly) is compact if $\lambda_n \to 0$.

Proof. Define $T_k : \ell^2 \to \ell^2$ by $T_k x := \sum_{n=1}^{k} \lambda_n (x, e_n) e_n$. Then each T_k is compact, and:

$$\begin{aligned}
\|(T - T_k)(x)\|_{\ell^2}^2 &= \left\| \sum_{n>k} \lambda_n (x, e^n) e^n \right\|_{\ell^2}^2 \\
&\leqslant \sum_{n>k} |\lambda_n|^2 |(x, e^n)|^2 \, \|e^n\|^2 \\
&= \sum_{n>k} |\lambda_n|^2 |(x, e^n)|^2 \\
&\leqslant \sum_{n>k} \|\lambda_n\|_{\ell^\infty}^2 \, |(x, e^n)|^2 \\
&= \|\lambda_n\|_{\ell^\infty}^2 \sum_{n>k} |(x, e^n)|^2 \\
&\leqslant \|\lambda_n\|_{\ell^\infty}^2 \, \|x\|_{\ell^2}^2 .
\end{aligned}$$

Square rooting both sides, then taking the supremum over all $x \in \ell^2$, $\|x\|_{\ell^2} = 1$ gives $\|T - T_k\| \leqslant \|\lambda_n\|_{\ell^\infty}$. Since $\lambda_n \to 0$, it must be the case that $\|T - T_k\| \to 0$. Hence $T_k \to T$. Since the set of all compact operators is closed in $\mathcal{L}(\ell^2)$, it must be the case that T is compact. □

Exercise 7. *Let X denote an inner-product space over $\mathbb{C}$, and let $T : X \to X$ be a linear operator which is self-adjoint. Prove the following facts concerning the eigenvalues, eigenspaces, and invariant subspaces of T:*

(a) *Show that all eigenvalues of T are real.*

(b) *Show that any two eigenvectors of T which correspond to distinct eigenvalues must be orthogonal.*

(c) *Suppose $Y \subset X$ is a T-invariant subspace of X. Show that $Y^{\perp}$ is also T-invariant.*

Proof. (a) Suppose $Tx = \lambda x$ for some nonzero $x \in X$. Since T is self-adjoint, we have $(Tx, x) = (x, Tx)$; i.e., $(\lambda x, x) = (x, \lambda x)$. This is equivalent to $\lambda(x, x) = \overline{\lambda}(x, x)$. Rearranging gives $(\lambda - \overline{\lambda})(x, x) = 0$. Since we assumed $x \neq 0$, it must be the case that $\lambda - \overline{\lambda} = 0$; i.e., $\lambda = \overline{\lambda}$. This can only be true if $\lambda \in \mathbb{R}$, hence any eigenvalue of T must be real.

(b) Let λ_1, λ_2 be distinct eigenvalues with corresponding eigenvectors u_1, u_2. Since T is self-adjoint, we have $(Tu_1, u_2) = (u_1, Tu_2)$. This is equivalent to $(\lambda_1 u_1, u_2) = (u_1, \lambda_2 u_2)$. Thus $\lambda_1(u_1, u_2) = \lambda_2(u_1, u_2)$. Rearranging gives $(\lambda_1 - \lambda_2)(u_1, u_2) = 0$. Since $\lambda_1 \neq \lambda_2$, it must be the case that $(u_1, u_2) = 0$. Thus $u_1 \perp u_2$.

(c) Let $z \in Y^{\perp}$. Then for any $y \in Y$ we have $(Tz, y) = (z, Ty)$. Since Y is T-invariant, z and Ty must be orthogonal, hence $(z, Ty) = 0$. This means $(Tz, y) = 0$ for all $y \in Y$, hence $Tz \in Y^{\perp}$. Thus $Y^{\perp}$ is also T-invariant. □

Week 8 Problems

Exercise 1. *With the notation we had in class: $X := \{f \in C^2([a, b]) \mid f$ satisfies BCs from RSL $\}$, and $Lu = -((pu')' + qu)$. Establish Lagrange's identity:*

$$uLv - vLu = -(p(uv' - vu'))'$$

Proof. Observe that:

$$\begin{aligned}
uLv - vLu &= -u((pv')' + qv) + v((pu')' + qu) \\
&= -u(p'v' + pv'' + qv) + v(p'u' + pu'' + qu) \\
&= -up'v' - upv'' - uqv + vp'u' + vpu'' + vqu \\
&= -up'v' - upv'' + vp'u' + vpu'' \\
&= -up'v' - upv'' - pu'v' + pu'v' + vp'u' + vpu'' \\
&= -(puv')' + (pvu')' \\
&= -(puv' - pvu')' \\
&= -(p(uv' - vu'))'.
\end{aligned}$$

□

Exercise 4. *Consider the RSL:*

$$X'' = -\lambda X, \quad 0 < x < 1, \quad X(0) = 0, \quad X'(1) = \alpha X(1),$$

where $\alpha > 0$. *Show, graphically, that is* α *is large enough then there will be exactly one negative eigenvalue. What value must* α *exceed to guarantee this?*

Proof. If $\lambda < 0$, take $\lambda := -\mu^2$. Our RSL becomes:

$$X'' - \mu X = 0.$$

The solution to this differential equation is:

$$X(x) = A\sinh(\mu x) + B\cosh(\mu x).$$

Plugging in our first boundary condition yields:

$$0 = X(0) = B.$$

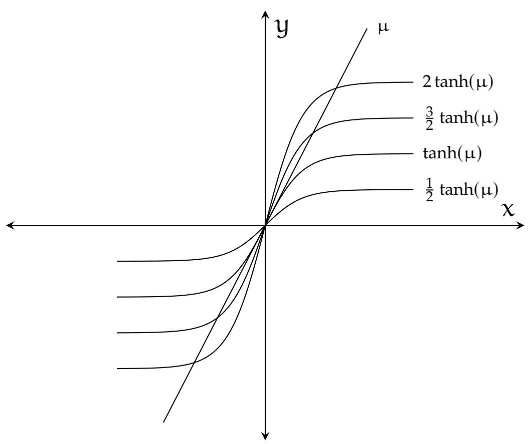
Now plugging in the second boundary condition gives:

$$\mu A\cosh(\mu) = \alpha A\sinh(\mu).$$

Simplifying and rearranging gives:

$$\mu = \alpha\tanh(\mu).$$

From the following graph, if $\alpha > 1$ then there will be exactly one negative eigenvalue.



□

Exercise 5. *Consider the RSL:*

$$X'' = -\lambda X, \quad 0 < x < 1, \quad X'(0) = aX(0), \quad X'(1) = -bX(1),$$

where $a, b \geqslant 0$*. Without solving for the eigenvalues show that all eigenvalues must be greater than or equal to 0. (Hint: Let* $L := -\frac{d^2}{dx^2}$*; assuming* $Lu = \lambda u$ *for a nonzero function* u *which satisfies the BCs, calculate* (Lu, u) *in two different ways.)*

Proof. Defining $L := -\frac{d^2}{dx^2}$, we have:

$$\begin{aligned}
(Lu, u) &= \int_0^1 -u''(x)\overline{u(x)}dx \\
&= -u'(x)\overline{u(x)}\Big|_0^1 + \int_0^1 u'(x)\overline{u'(x)}dx \\
&= -u'(1)\overline{u(1)} + u'(0)\overline{u(0)} + \int_0^1 u'(x)\overline{u'(x)}dx \\
&= bu(1)\overline{u(1)} + au(0)\overline{u(0)} + \int_0^1 u'(x)\overline{u'(x)}dx \\
&= b|u(1)|^2 + a|u(0)|^2 + \int_0^1 |u'(x)|^2 dx.
\end{aligned}$$

Similarly:

$$\begin{aligned}
(\lambda u, u) &= \int_0^1 \lambda u(x)\overline{u(x)}dx \\
&= \lambda \int_0^1 |u(x)|^2 dx.
\end{aligned}$$

Since $Lu = \lambda u$, we have $(Lu, u) = (\lambda u, u)$. Solving for λ gives:

$$\lambda = \frac{b|u(1)|^2 + a|u(0)|^2 + \int_0^1 |u'(x)|^2 dx}{\int_0^1 |u(x)|^2 dx}.$$

Since $u(x)$ is a nonzero function, the right hand side of the above equation is not undefined. Moreover, each term in the numerator is greater than or equal to zero, hence $\lambda \geqslant 0$. □

Exercise 6. *Let* k *be a continuous function on* $[a,b]\times[a,b]$*. Define the operator* $K : L^2([a,b]) \to L^2([a,b])$ *by:*

$$(Kg)(x) = \int_a^b k(x,t)g(t)dt.$$

Use Cauchy-Schwartz and the continuity of k *to show that* Kg *is continuous for any* $g \in L^2([a,b])$*.*

Proof. Let $g \in L^2([a,b])$. Let $x \in [a,b]$ and $\epsilon > 0$. Since k is continuous, there exists a $\delta > 0$ such that, for all $y \in [a,b]$, we have $|x-y| < \delta$ implies:

$$|k(x,t) - k(y,t)| < \frac{\epsilon}{\|g\|_{L^2}}.$$

For any $y \in [a,b]$, if $|x-y| < \delta$ then:

$$\begin{aligned}
|(Kg)(x) - (Kg)(y)| &\leqslant \int_a^b |k(x,t) - k(y,t)| \cdot |g(t)|dt \\
&\leqslant \left(\int_a^b |k(x,t) - k(y,t)|^2 dt\right)^{\frac{1}{2}} \cdot \|g\|_{L^2} \\
&< \left(\int_a^b \frac{\epsilon^2}{\|g\|_{L^2}^2} dt\right)^{\frac{1}{2}} \|g\|_{L^2} \\
&= \frac{\epsilon}{\|g\|_{L^2}} \|g\|_{L^2} \\
&= \epsilon.
\end{aligned}$$

Thus Kg is continuous. □